

2024-25 - FIRST SEMESTER
FINANCIAL TIME SERIES (SFA01)
MASTER 2, PRINCIPAL SESSION

Written exam of January 28, 2025

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Durée: 2 heures.

Authorized documents: handwritten double-sided sheet on A4 paper.

Duration: 2 hours.

Preamble for all exercises

Reminders on projections in Hilbert spaces: Let H be an Hilbert space (e.g. $L^2(\Omega, \mathcal{F}, \mathbb{P})$) and let E and F be two closed subspaces of H . If $E \perp F$, we have, for all $x \in H$,

$$\text{proj} \left(x \mid E \oplus F \right) = \text{proj} (x \mid E) + \text{proj} (x \mid F) .$$

If $E \subseteq F$, we have, for all $x \in H$,

$$\text{proj} (x \mid E) = \text{proj} (\text{proj} (x \mid F) \mid E) .$$

Some processes We will consider the following processes:

1. $V = (V_t)_{t \in \mathbb{Z}}$ and $W = (W_t)_{t \in \mathbb{Z}}$ are two uncorrelated real valued centered white noises with respective variances ρ^2 and σ^2 .
2. $X = (X_t)_{t \in \mathbb{Z}}$ is a real valued AR(1) process with canonical representation

$$X_t = \phi X_{t-1} + W_t, \quad t \in \mathbb{Z}, \quad (1)$$

where ϕ is a real parameter.

3. We define for all $t \in \mathbb{Z}$,

$$Y_t = X_t + V_t. \quad (2)$$

Some notation Recall that for a L^2 process $Y = (Y_t)_{t \in \mathbb{Z}}$ valued in $\mathbb{R}$ or $\mathbb{C}$ and defined on $(\Omega, \mathcal{F}, \mathbb{P})$, we denote $\mathcal{H}_{t,p}^Y = \text{Span} (Y_s : t-p < s \leq t)$ and $\mathcal{H}_t^Y = \overline{\text{Span}} (Y_s : s \leq t)$, seen as two subspaces of the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$.

Additional notation From the previous point, we have, for all $t \geq 1$, $\mathcal{H}_{t,t}^Y = \text{Span} (Y_s : 1 \leq s \leq t)$. We further define, for all $s \in \mathbb{Z}$ and $t \geq 1$,

$$\hat{X}_{s|t} = \text{proj} (X_s \mid \mathcal{H}_{t,t}^Y), \quad (3)$$

$$\sigma_{s|t}^2 = \mathbb{E} \left[(X_s - \hat{X}_{s|t})^2 \right]. \quad (4)$$

Exercises

Exercise 1. We denote for all $t \in \mathbb{Z}$, $\vec{U}_t = [X_t \ Y_t]^T$, which is thus an $\mathbb{R}^2$ -valued process.

1. Justify that $\vec{U} = (\vec{U}_t)_{t \in \mathbb{Z}}$ is a VAR(1) process and provide a reduced form representation of this process, that is, an equation of the form

$$\vec{U}_t = \Phi \vec{U}_{t-1} + \vec{Z}_t, \quad t \in \mathbb{Z},$$

where $\vec{Z} = (\vec{Z}_t)_{t \in \mathbb{Z}}$ is a centered white noise.

2. Give the covariance matrix Σ of $\vec{Z}$ and the spectral density matrix function of $\vec{U}$.
3. Deduce from Σ whether X instantaneously Granger-causes Y and whether Y instantaneously Granger-causes X .
4. What is the nature of the process Y ?

Exercise 2. We investigate some questions related to the Kalman algorithm.

5. Justify that $|\phi| < 1$ and that $\mathcal{H}_t^X = \mathcal{H}_t^W$ for all $t \in \mathbb{Z}$.
6. Compute the mean and variance of X_0 .

From now on we suppose that we observe $Y_1, \dots, Y_n$, for some $n \geq 1$. In this setting, assuming that the parameters ϕ, σ^2 and ρ^2 are known, the Kalman algorithm provides a way to compute $\hat{X}_{t|t}$ and $\sigma_{t|t}^2$ for all $t = 0, \dots, n$ online, using the initial conditions $\hat{X}_{0|0} = \mathbb{E}[X_0]$ and $\sigma_{0|0}^2 = \text{Var}(X_0)$ and the following iterative formulas for $t = 0, \dots, n-1$:

$$\begin{aligned} \hat{X}_{t+1|t} &= \phi \hat{X}_{t|t}, \\ \sigma_{t+1|t}^2 &= \phi^2 \sigma_{t|t}^2 + \sigma^2, \\ k_{t+1} &= \left(\sigma_{t+1|t}^2 + \rho^2 \right)^{-1} \sigma_{t+1|t}^2, \\ \hat{X}_{t+1|t+1} &= \hat{X}_{t+1|t} + k_{t+1} (Y_{t+1} - \hat{X}_{t+1|t}), \\ \sigma_{t+1|t+1}^2 &= (1 - k_{t+1}) \sigma_{t+1|t}^2. \end{aligned}$$

7. How many operations are required to obtain the quantities $\hat{X}_{t|t}$, $\sigma_{t|t}^2$, $\hat{X}_{t+1|t}$ and $\sigma_{t+1|t}^2$ for all $t = 1, \dots, n$? (give only the main order of magnitude in the form $O(n^\alpha)$)
8. Using the evolution equation (1), show that, for all $h \geq 1$,

$$\hat{X}_{t+h+1|t} = \phi \hat{X}_{t+h|t} \quad \text{and} \quad \sigma_{t+h+1|t}^2 = \phi^2 \sigma_{t+h|t}^2 + \sigma^2$$

Exercise 3. From now on, we let $1 \leq s \leq n-1$ and denote

$$\mathcal{V}_{s,n} = \mathcal{H}_{s,s}^Y + \text{Span}(X_{s+1} - \hat{X}_{s+1|s}) + \text{Span}(W_t : s+1 < t \leq n) + \text{Span}(V_t : s+1 \leq t \leq n).$$

We further define

$$\tilde{X}_{s|n} = \text{proj}(X_s | \mathcal{V}_{s,n}).$$

9. Show that $\mathcal{H}_{n,n}^Y \subset \mathcal{V}_{s,n}$.
10. Show that
- $$\tilde{X}_{s|n} = \hat{X}_{s|s} + j_s \left(X_{s+1} - \hat{X}_{s+1|s} \right) \quad \text{with} \quad j_s = \phi \sigma_{s+1|s}^{-2} \sigma_{s|s}^2 .$$
11. Deduce a formula for $\hat{X}_{s|n}$ using $\hat{X}_{s|s}$, $\hat{X}_{s+1|n}$ and $\hat{X}_{s+1|s}$.
12. Show that $X_s - \hat{X}_{s|n}$ is orthogonal to $\hat{X}_{s+1|n} - \hat{X}_{s+1|s}$ and deduce a formula for $\sigma_{s|n}^2$ using $\sigma_{s|s}^2$ and $\text{Var} \left(\hat{X}_{s+1|n} - \hat{X}_{s+1|s} \right)$.
13. Use that $X_{s+1} - \hat{X}_{s+1|n}$ is orthogonal to $\hat{X}_{s+1|n} - \hat{X}_{s+1|s}$ to compute $\text{Var} \left(\hat{X}_{s+1|n} - \hat{X}_{s+1|s} \right)$ with $\sigma_{s+1|s}^2$ and $\sigma_{s+1|n}^2$.
14. Conclude that
- $$\sigma_{s|n}^2 = \sigma_{s|s}^2 + \phi^2 \sigma_{s+1|s}^{-4} \sigma_{s|s}^4 \left(\sigma_{s+1|n}^2 - \sigma_{s+1|s}^2 \right) . \quad (5)$$
15. Describe a complete algorithm to compute $\hat{X}_{s|n}$ and $\sigma_{s|n}^2$ for all $1 \leq s \leq n$ from the observations $Y_1, \dots, Y_n$ and the parameters ϕ, σ^2 and ρ^2 .

Exercise 4. From now on, we set, for all $k \in \mathbb{Z}$,

$$v_k = \lim_{s \rightarrow \infty} \sigma_{s|s+k}^2 .$$

16. Using the definitions in Eq. (3) and (4), justify that, for any given $k \in \mathbb{Z}$, the above limit is well defined in $\mathbb{R}_+$.
17. Deduce an iterative formula for the sequence $(v_k)_{k \geq 1}$ from (5).
18. Deduce the asymptotic behavior of v_k as $k \rightarrow \infty$. Is it an increasing or decreasing sequence ?

Answers

Solution of Exercise 1 1.

2.

3.

4.

Solution of Exercise 2 5.

6.

7.

8.

Solution of Exercise 3 9.

10.

11. Since $\mathcal{H}_{n,n}^Y \subset \mathcal{V}_{s,n}$, we have

$$\hat{X}_{s|n} = \text{proj} (X_s | \mathcal{H}_{n,n}^Y) = \text{proj} (\text{proj} (X_s | \mathcal{V}_{s,n}) | \mathcal{H}_{n,n}^Y) = \text{proj} (\tilde{X}_{s,n} | \mathcal{H}_{n,n}^Y) .$$

From the previous expression of $\tilde{X}_{s,n}$, we get that

$$\hat{X}_{s|n} = \hat{X}_{s|s} + j_s (\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) .$$

12. By definition, we have that $X_s - \hat{X}_{s|n} \perp \mathcal{H}_{n,n}^Y$ and thus is orthogonal to $\hat{X}_{s+1|n}$ and $\hat{X}_{s+1|s}$ which both belong to $\mathcal{H}_{n,n}^Y$ (since $1 \leq s < n$). Using that

$$X_s - \hat{X}_{s|s} = (X_s - \hat{X}_{s|n}) + j_s (\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) ,$$

we obtain that

$$\sigma_{s|s}^2 = \sigma_{s|n}^2 + j_s^2 \text{Var} (\hat{X}_{s+1|n} - \hat{X}_{s+1|s})$$

Hence $\sigma_{s|n}^2 = \sigma_{s|s}^2 - j_s^2 \text{Var} (\hat{X}_{s+1|n} - \hat{X}_{s+1|s})$.

13. We have

$$(\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) + (X_{s+1} - \hat{X}_{s+1|n}) = X_{s+1} - \hat{X}_{s+1|s}$$

And using that $\hat{X}_{s+1|n} - \hat{X}_{s+1|s} \in \mathcal{H}_n^Y \perp X_{s+1} - \hat{X}_{s+1|n}$ we get that

$$\text{Var} (\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) + \text{Var} (X_{s+1} - \hat{X}_{s+1|n}) = \text{Var} (X_{s+1} - \hat{X}_{s+1|s}) ,$$

or, equivalently, $\text{Var} (\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) + \sigma_{s+1|n}^2 = \sigma_{s+1|s}^2$, that is, finally,

$$\text{Var} (\hat{X}_{s+1|n} - \hat{X}_{s+1|s}) = \sigma_{s+1|s}^2 - \sigma_{s+1|n}^2 .$$

14. The two previous questions give us that

$$\sigma_{s|n}^2 = \sigma_{s|s}^2 - j_s^2 \left(\sigma_{s+1|s}^2 - \sigma_{s+1|n}^2 \right)$$

Replacing j_s with its definition, we get the given formula.

15. (...)

Solution of Exercice 4 16. By weak stationarity we have that, for all $k \in \mathbb{Z}$,

$$\sigma_{s|s+k}^2 = \text{Var} \left(X_0 - \text{proj} \left(X_0 | \mathcal{H}_{k,s+k}^Y \right) \right)$$

Hence, since $\left(\mathcal{H}_{k,s+k}^Y \right)_{s \geq 1}$ is an increasing sequence of closed vector subspaces whose union's closure is $\mathcal{H}_k^Y$, we have

$$\lim_{s \rightarrow \infty} \sigma_{s|s+k}^2 = \text{Var} \left(X_0 - \text{proj} \left(X_0 | \mathcal{H}_k^Y \right) \right) .$$

17. Taking $n = s + k$ and then the limit as $s \rightarrow \infty$ in the formula of Question 14 we have

$$v_k = v_0 + \phi^2 v_{-1}^{-2} v_0^2 (v_{k-1} - v_{-1}) .$$

18. Let $\ell = (1 - \phi^2 v_{-1}^{-2} v_0^2)^{-1} (v_0 - \phi^2 v_{-1}^{-1} v_0^2)$ so that

$$\ell = v_0 + \phi^2 v_{-1}^{-2} v_0^2 (\ell - v_{-1}) .$$

and taking the difference with the iterative formula,

$$(v_k - \ell) = \phi^2 v_{-1}^{-2} v_0^2 (v_{k-1} - \ell) .$$

Moreover we have using that $\sigma_{t+1|t}^2 = \phi^2 \sigma_{t|t}^2 + \sigma^2$, and letting $t \rightarrow \infty$, $v_{-1} = \phi^2 v_0 + \sigma^2$. Thus we get

$$\phi^2 v_{-1}^{-2} v_0^2 = \frac{\phi^2 v_0}{\phi^2 v_0 + \sigma^2} < 1 .$$

We deduce that (v_k) converges geometrically fast to ℓ . Moreover since $v_k = \text{Var} \left(X_0 - \text{proj} \left(X_0 | \mathcal{H}_k^Y \right) \right)$, and $(\mathcal{H}_k^Y)$ is an increasing sequence of closed vector subspaces, we know that it decreases